

DATABASE DESIGN

Assignment: From ERD to SQL Tables

Conceptual Design • Relational Mapping • Physical Implementation

Assignment Goal

Translate a real-world business case into a correct conceptual ERD, map the ERD into a relational schema, and implement the final design using SQL CREATE TABLE statements with appropriate integrity constraints.

Learning Outcomes

- Identify entities, attributes, candidate keys, primary keys, and relationship types from written requirements.
- Represent cardinality and participation constraints correctly in an ERD.
- Resolve many-to-many relationships using associative entities during relational mapping.
- Convert a conceptual model into relations with primary and foreign keys.
- Create database tables using suitable data types and integrity constraints.

General Instructions

- Use Draw.io or any similar tool.
- Your ERD must show entity names, attributes, primary keys, relationships, cardinalities, and optional/mandatory participation.
- State any reasonable assumption that is not explicitly defined in the problem.
- The relational mapping must be derived from your ERD. Do not redesign the database independently during Task 2.
- For SQL implementation, use the DBMS syntax taught in class. Your script must execute in dependency order without foreign-key errors.
- Submit the ERD, relational mapping, assumptions, and SQL script as one complete assignment.

Problem 1 — FitZone Fitness Center

FitZone Fitness Center has decided to replace its spreadsheets with a relational database. The center wants to manage members, membership plans, trainers, fitness classes, rooms, and trainer certifications. You have been hired as the database designer.

- Each member has a unique Member ID, first name, last name, date of birth, phone number, and an address consisting of street and city.
- Each membership plan has a unique Plan ID, a plan name, duration in months, and a monthly fee.
- Every member must subscribe to exactly one membership plan. A membership plan may be used by many members.
- Each trainer has a unique Trainer ID, first name, last name, phone number, hire date, and a specialty.
- Each fitness class has a unique Class ID, class name, difficulty level, class date, start time, and maximum capacity.
- Each fitness class is led by exactly one trainer. A trainer may lead several fitness classes, or may currently lead none.
- Each room has a unique Room Number, room name, and capacity.
- Each fitness class is held in exactly one room. A room may be used for many fitness classes at different times.
- A member may register for many fitness classes, and each fitness class may have many registered members.
- For every member registration in a fitness class, FitZone must store the Registration Date and Registration Status (for example: Registered, Attended, Cancelled).
- Each certification has a unique Certification Code, certification name, and issuing organization.
- A trainer may hold several certifications, and the same certification may be held by several trainers. FitZone also wants to store the date on which a trainer earned each certification.

Design Note: Some requirements describe many-to-many relationships with attributes of their own. Your final relational model must preserve those attributes correctly.

Task 1 — Conceptual ERD Design

Weight: 40 marks

Create a complete conceptual ERD for FitZone. Your ERD must include:

- All entities required by the business rules.
- All relevant attributes for every entity.
- Primary key identification for each strong entity.
- Composite attributes where appropriate (for example, address can be shown as Street + City).
- All relationships with meaningful relationship names.
- Minimum and maximum cardinality for both sides of every relationship.
- Mandatory versus optional participation.
- Correct treatment of any M:N relationship and any relationship that has its own attributes.
- A short list of assumptions made while interpreting the requirements.

Deliverable 1: One clearly labeled ERD + assumptions.

Task 2 — Relational Mapping

Weight: 30 marks

Map your final ERD into a relational schema. For each relation, write the table name followed by its attributes and clearly identify keys.

- Map every regular entity into a relation.
- Map each 1:M relationship by placing the foreign key on the correct side.
- Resolve each M:N relationship using an associative relation/table.
- Carry relationship attributes into the correct associative relation.
- Identify every Primary Key (PK).
- Identify every Foreign Key (FK), including the referenced relation and referenced key.
- Indicate composite primary keys where your design uses them.
- Ensure the mapping preserves all cardinality rules from Task 1.

Required mapping format — syntax example only

ENTITY_NAME(**PrimaryKey**, Attribute1, Attribute2, ForeignKey)

FK ForeignKey → PARENT_TABLE(PrimaryKey)

Deliverable 2: A complete relational schema matching your ERD.

Task 3 — SQL Table Creation

Weight: 30 marks

Implement the relational schema from Task 2 using SQL DDL. Your script must contain CREATE TABLE statements for all relations in your design.

- Choose an appropriate SQL data type for every column.
- Define a PRIMARY KEY constraint for every table.
- Define all required FOREIGN KEY constraints.
- Use NOT NULL for mandatory attributes.
- Use UNIQUE where a business rule requires uniqueness beyond the primary key.
- Use CHECK constraints for suitable domain rules (for example, positive fees/capacities or an allowed set of statuses).
- Create parent tables before child tables so that the script can execute successfully.
- Use clear and consistent table/column naming conventions.
- Add short SQL comments where a design choice may not be immediately obvious.

Minimum Integrity Rules Expected

Constraint	What it protects	Expected use
PRIMARY KEY	Entity uniqueness	Every table
FOREIGN KEY	Referential integrity	All mapped relationships
NOT NULL	Mandatory data	Required attributes / mandatory FK
UNIQUE	Alternate uniqueness	Where business rules require it
CHECK	Valid domain values	Fees, capacity, status, duration, etc.

Deliverable 3: One executable .sql script containing all CREATE TABLE statements and constraints.

Marking Rubric

Area	Marks	Evaluation focus
Task 1: Conceptual ERD	40	Correct entities/attributes, keys, relationships, cardinalities, participation, assumptions
Task 2: Relational Mapping	30	Correct relations, PK/FK placement, M:N resolution, preservation of relationship attributes
Task 3: SQL DDL	30	Correct data types, CREATE TABLE order, PK/FK, nullability, uniqueness, checks, consistency
Total	100	Complete and internally consistent database design